

Software architecture course project

Workout progress tracker

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Project description

People that workout encounter a problem when you need to document your personal records. Information, such as lifted weight, repetition or set count, helps to track your individual progress. This mobile application creates an easy-to-use UI that allows you to document such information quickly and without much hassle. Additionally, the application helps you track your progress over a period of time.

Technologies used

React Native, Expo, Nativewind

React native – technology that allows you to create mobile applications for iOS and Android without the need to change the codebase.

Expo – framework that makes mobile development faster and more convenient.

Nativewind – framework that adds default responsiveness to your application.

Laravel

Laravel – already familiar technology therefore, the development of the backend should be way faster. It also gives you the possibility to add a web application without much additional work to the backend.

PostgreSQL

PostgreSQL – also a familiar technology and integrates well with Laravel.

Functionalities

- First iteration
 - User registration.
 - User session.
- Second iteration
 - Exercises CRUD.
 - Workout CRUD.
- Third iteration
 - Workout session CRUD.
 - Workout session and related data tracking.
 - Admin GUI (Web based interface, allows administration to insert public exercises and workouts).
- Fourth iteration
 - Local data caching without internet connection and data synchronization with connection.
 - Progress analysis charts based on criteria (based on lifted weight, based on repetitions done).

Non-functional requirements

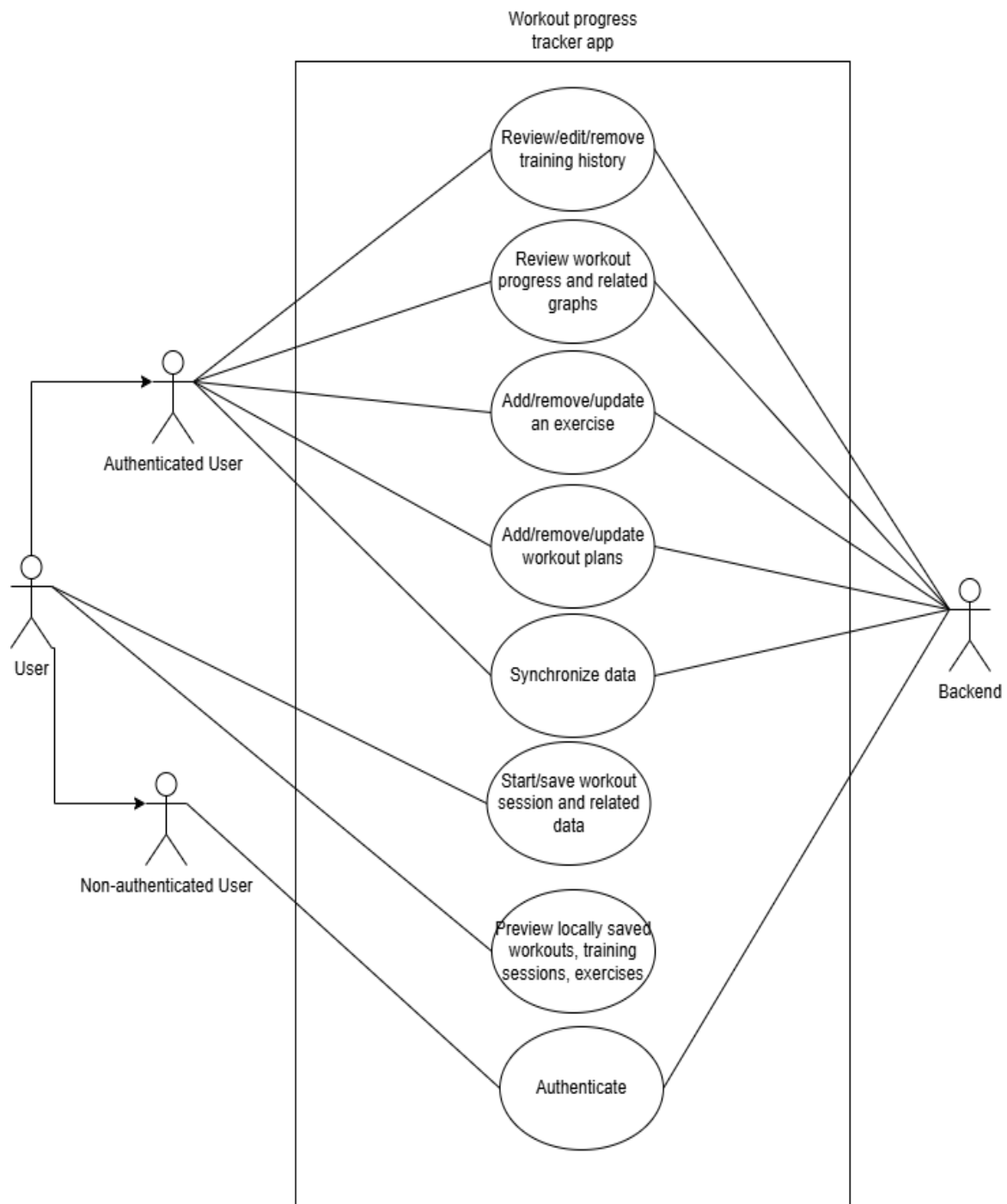
Security – the application must be secure as all the data will be passed through the network.

Usability - the application must be easy and fast to use as the main use case is to track your data during a workout.

Scalability – program should be easy to scale up, this is one of the reasons why Laravel was used.

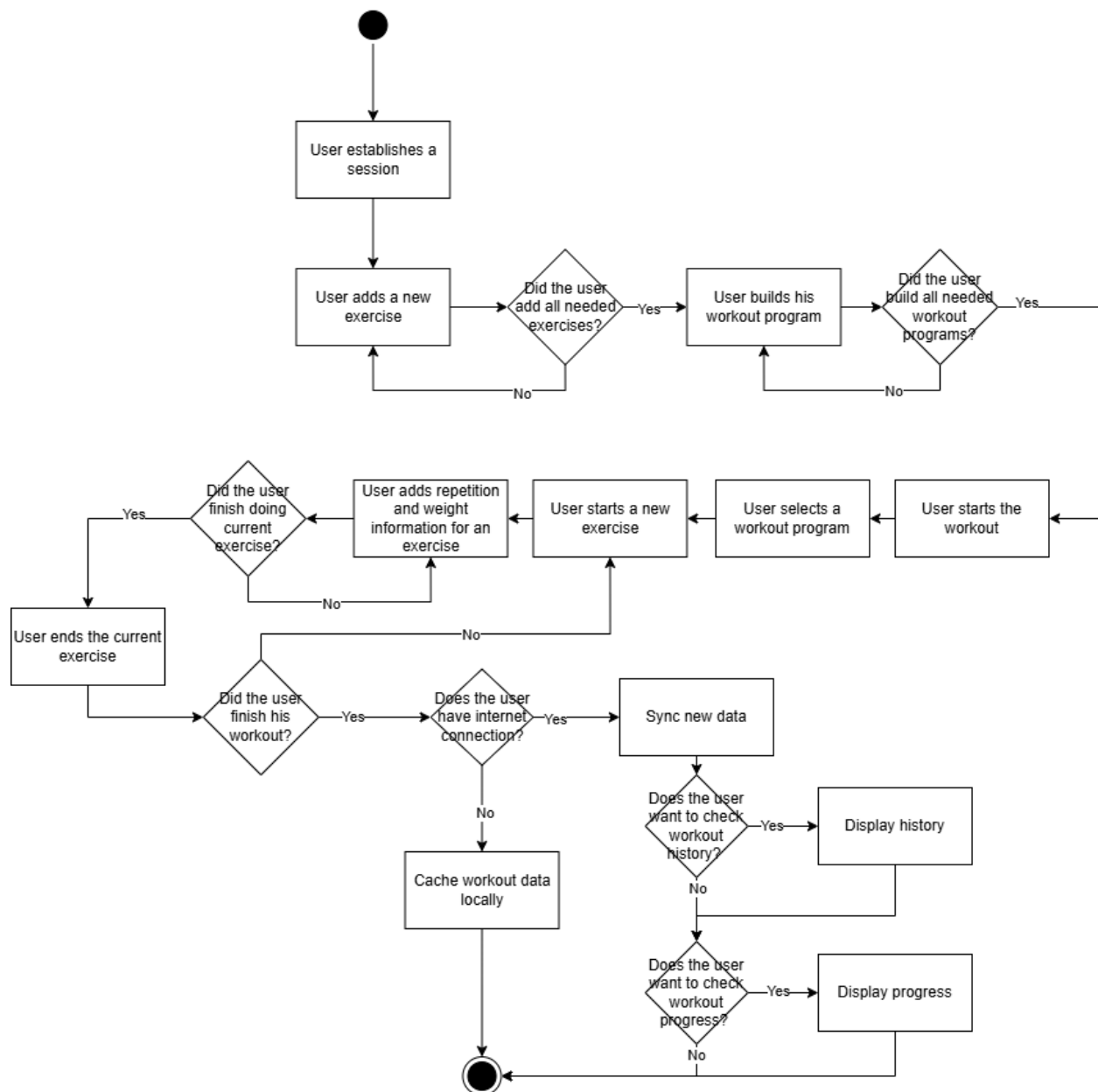
UML diagrams

Use case diagram



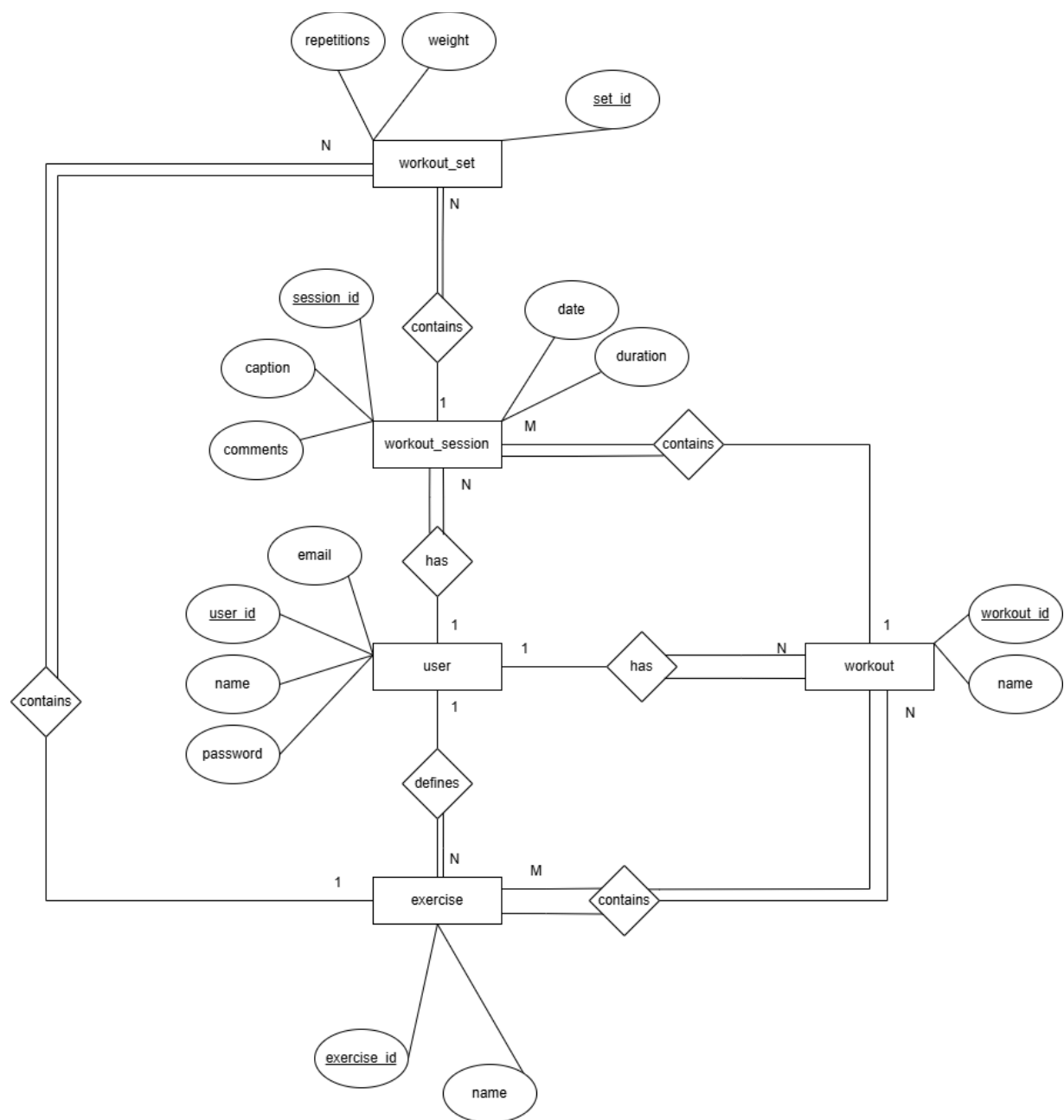
The diagram displays functionalities of two user types, authenticated user and non-authenticated user. Authenticated users have full capabilities of the application, while non-authenticated users have capabilities that only involve local data.

Activity diagram



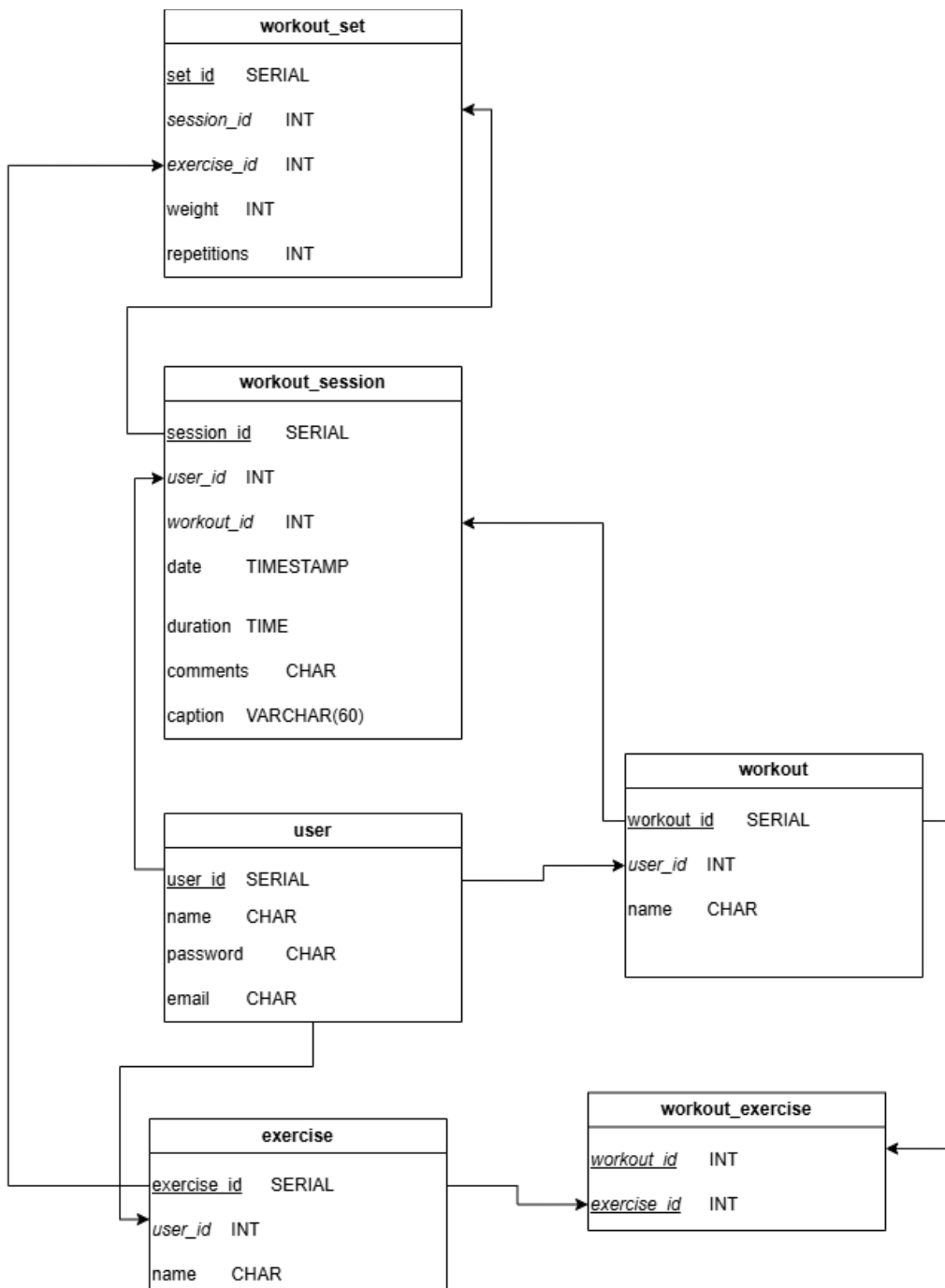
The activity diagram is displayed from a first-time authenticated user perspective. Users have the ability to create their own exercises as well as workouts. The reasoning behind this decision and not using templates is that all workouts can be very different and include many different exercises. Therefore, it is nearly impossible to suit every user. Once the user builds his workout routines, he can start workout tracking, select a workout from his presets, and all that is left to do is to document the data. After tracking the application makes next actions based on whether there is an internet connection or not.

ER model diagram



The diagram sketches the preliminary example of the database tables. The user can create as many as he wants exercises and workouts, he can have many workout sessions. Exercise can repeat in different workouts as well as in workout sets. Workout contains many different exercises. Workout sessions can be with the same workout many different times, also it usually contains more than one set per exercise.

DB schema



Summarizes the previous diagram by placing FKs in the right tables, also shows the variable types in each table.